

## 32.1 Random effects

A one-way layout (Definition 15.1.1) has  $g$  groups and  $g$  group means. In Part IV those means were the object of the study: the five nitrogen rates were chosen, and every conclusion was about them. [Section 15.1](#) previewed the other possibility and left it here; this section carries the preview through. Consider a study of a different shape. A reference material with a known lead content is sent to twelve accredited laboratories, each of which analyses four aliquots of it. Nobody cares whether laboratory 3 reads higher than laboratory 7. The question is how much accredited laboratories disagree with one another, because that number, and not the twelve particular deviations, is what a client using an unnamed laboratory faces. The twelve are a sample from a population of laboratories, and the conclusion must reach the population.

### 32.1.1 When a factor is random

The test, met first in [Section 15.1](#), is about the *scope of the conclusion*. If the levels in the data exhaust the levels you want to talk about, the factor is fixed; if they are a sample and the conclusion should extend to levels you did not observe, it is random. Doses, treatments, varieties and the two arms of a trial are almost always fixed; laboratories, batches of reagent, subjects, litters, classrooms and interviewers are usually random. Three things change when a factor is declared random.

- (1) **The parameters change.** The  $g$  group effects are replaced by a single number, their variance. A study with  $g = 12$  laboratories estimates one variance, not twelve means, and its precision is governed by  $g$ , not by the total number of observations.
- (2) **Estimation becomes prediction.** The realized effect of laboratory 3 is a random variable, not a parameter. Asking for its value is a prediction problem, and [Section 32.3](#) shows that the best answer is not the laboratory's own average.
- (3) **The covariance changes.** Two aliquots analysed by the same laboratory share its effect, so they are correlated. The model is no longer a linear model with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , and this is the reason the chapter belongs to Part VII.

*Warning.* A variance is estimated from the number of *levels*, not the number of observations. With  $g = 3$  laboratories there are two degrees of freedom for the between-laboratory variance, and any interval for it is nearly useless however many aliquots each laboratory analyses. A treated and a control arm should never be a random factor: two levels carry no information about a population of levels, and there is no population. When a factor has few levels but its levels really are a sample, the honest options are to report the fixed-effect analysis alongside, or to supply prior information ([Section 32.6](#)).

### 32.1.2 The one-way random effects model

#### Definition 32.1.1. One-way random effects model

Let  $g \geq 2$  groups contain  $n_k \geq 1$  observations each, with  $n = \sum_k n_k$ . The **one-way random effects model** is

$$Y_{kj} = \mu + a_k + \varepsilon_{kj}, \quad k = 1, \dots, g, \quad j = 1, \dots, n_k,$$

where  $\mu$  is an unknown constant, the **group effects**  $a_1, \dots, a_g$  are uncorrelated with mean 0 and variance  $\sigma_a^2 \geq 0$ , the errors  $\varepsilon_{kj}$  are uncorrelated with mean 0 and variance  $\sigma^2 > 0$ , and the two sets are uncorrelated with each other. In matrix form, with  $\mathbf{Z}$  the  $n \times g$  matrix of group indicators,

$$\mathbf{Y} = \mu \mathbf{1} + \mathbf{Z}\mathbf{a} + \boldsymbol{\varepsilon}.$$

The **normal** one-way random effects model adds  $\mathbf{a} \sim N_g(\mathbf{0}, \sigma_a^2 \mathbf{I})$  independent of  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ . The layout is **balanced** if  $n_1 = \dots = n_g = m$ . The pair  $(\sigma_a^2, \sigma^2)$  are the **variance components**.

The model has three parameters whatever  $g$  is, where Definition 15.1.1 had  $g + 1$ . That is the economy of random effects, and also their risk: it is bought by assuming that the  $g$  effects are an uncorrelated sample from one distribution.

### Proposition 32.1.1. The one-way random effects model

In Definition 32.1.1, write  $\gamma = \sigma_a^2/\sigma^2$  and let  $\mathbf{M}$  be the projection onto  $\mathcal{C}(\mathbf{Z})$ , which replaces each entry by its group mean.

- (a)  $E(\mathbf{Y}) = \mu \mathbf{1}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ , a block diagonal matrix whose  $k$ th block is  $\sigma^2 \mathbf{I}_{n_k} + \sigma_a^2 \mathbf{1}\mathbf{1}^\top$ .
- (b) (*Intraclass correlation.*) Observations in different groups are uncorrelated, and two observations in the same group have correlation

$$\rho = \frac{\sigma_a^2}{\sigma_a^2 + \sigma^2} = \frac{\gamma}{1 + \gamma} \in [0, 1).$$

- (c) (*Moment estimates.*) Let MSB and MSE be the between and within mean squares of the one-way analysis of variance, on  $g - 1$  and  $n - g$  degrees of freedom. Then  $E(\text{MSE}) = \sigma^2$  and  $E(\text{MSB}) = \sigma^2 + m_0 \sigma_a^2$  with

$$m_0 = \frac{1}{g - 1} \left( n - \frac{\sum_k n_k^2}{n} \right), \quad (32.1.1)$$

which equals  $m$  in a balanced layout. Hence  $\hat{\sigma}^2 = \text{MSE}$  and  $\hat{\sigma}_a^2 = (\text{MSB} - \text{MSE})/m_0$  are unbiased.

- (d) (*Distributions, balanced and normal.*) If the layout is balanced and the model is normal, then with  $\lambda_1 = \sigma^2 + m\sigma_a^2$ ,

$$\frac{\text{SSB}}{\lambda_1} \sim \chi^2(g - 1) \quad \text{independently of} \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - g),$$

and therefore  $\text{MSB}/\text{MSE} \sim (1 + m\gamma) F(g - 1, n - g)$ .

*Proof.*

- (a) By Theorem 2.2.1 applied to  $\mathbf{Y} = \mu \mathbf{1} + [\mathbf{Z}, \mathbf{I}](\mathbf{a}^\top, \boldsymbol{\varepsilon}^\top)^\top$ ,  $\text{Cov}(\mathbf{Y}) = \mathbf{Z}(\sigma_a^2 \mathbf{I}_g) \mathbf{Z}^\top + \sigma^2 \mathbf{I}$ . The matrix  $\mathbf{Z}\mathbf{Z}^\top$  has  $(i, i')$  entry 1 when observations  $i$  and  $i'$  are in the same group and 0 otherwise, which is the block form stated.
- (b) Read off (a): the covariance of two distinct observations in group  $k$  is  $\sigma_a^2$  and each variance is  $\sigma_a^2 + \sigma^2$ .
- (c) The two sums of squares are  $\text{SSB} = \mathbf{Y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{Y}$  and  $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ , where

$\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$ . Both projections annihilate  $\mathbf{1}$ , so Theorem 2.3.1 leaves only the trace term. Since the columns of  $\mathbf{Z}\mathbf{Z}^\top$  lie in  $\mathcal{C}(\mathbf{Z})$  we have  $\mathbf{M}\mathbf{Z}\mathbf{Z}^\top = \mathbf{Z}\mathbf{Z}^\top$ , and  $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ , so

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - g),$$

while  $\text{tr}(\mathbf{Z}\mathbf{Z}^\top) = n$  and  $\text{tr}(\mathbf{P}_1\mathbf{Z}\mathbf{Z}^\top) = n^{-1}\mathbf{1}^\top\mathbf{Z}\mathbf{Z}^\top\mathbf{1} = n^{-1}\sum_k n_k^2$  give

$$E(\text{SSB}) = \sigma^2(g - 1) + \sigma_a^2\left(n - \frac{\sum_k n_k^2}{n}\right).$$

Dividing by the degrees of freedom gives the two expectations, and unbiasedness of the two estimates follows by linearity. In a balanced layout  $n = gm$  and  $\sum_k n_k^2 = gm^2$ , so  $m_0 = \{gm - gm^2/(gm)\}/(g - 1) = m(g - 1)/(g - 1) = m$ .

(d) In the balanced case  $\mathbf{Z}\mathbf{Z}^\top = m\mathbf{M}$ , so by (a)

$$\text{Cov}(\mathbf{Y}) = \lambda_1\mathbf{M} + \sigma^2(\mathbf{I} - \mathbf{M}), \quad (32.1.2)$$

a spectral decomposition into the two orthogonal pieces  $\mathcal{C}(\mathbf{Z})$  and  $\mathcal{C}(\mathbf{Z})^\perp$ . Write  $\mathbf{\Sigma} = \text{Cov}(\mathbf{Y})$ . For the between projection  $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$ , which satisfies  $\mathbf{A}\mathbf{M} = \mathbf{A}$  and  $\mathbf{A}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , we get  $\mathbf{A}\mathbf{\Sigma} = \lambda_1\mathbf{A}$ , so  $\mathbf{A}\mathbf{\Sigma}/\lambda_1 = \mathbf{A}$  is idempotent of rank  $g - 1$  and  $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$ ; Theorem 4.3.2 gives  $\text{SSB}/\lambda_1 \sim \chi^2(g - 1)$ . For  $\mathbf{B} = \mathbf{I} - \mathbf{M}$  we get  $\mathbf{B}\mathbf{\Sigma} = \sigma^2\mathbf{B}$  and  $\text{SSE}/\sigma^2 \sim \chi^2(n - g)$  in the same way. Independence follows from Theorem 4.4.2, because  $\mathbf{A}\mathbf{\Sigma}\mathbf{B} = \lambda_1\mathbf{A}\mathbf{B} = \mathbf{0}$ . The ratio of the two mean squares is then  $(\lambda_1/\sigma^2)$  times an  $F(g - 1, n - g)$  variable, and  $\lambda_1/\sigma^2 = 1 + m\gamma$ . ■

Part (c) is Exercise (B1) carried to unbalanced layouts, and part (d) is the fact that makes the balanced case so pleasant: the same  $F$  ratio that tests equality of fixed group means tests  $\sigma_a^2 = 0$ , and it does more, because its distribution is known for every  $\gamma$ , not only at  $\gamma = 0$ . Inverting it gives an exact interval.

### Corollary 32.1.2. Exact interval for the variance ratio

In the balanced normal one-way random effects model, let  $F = \text{MSB}/\text{MSE}$  and let  $F_q$  denote the  $q$  quantile of  $F(g - 1, n - g)$ . Then

$$\left[ \frac{1}{m} \left( \frac{F}{F_{1-\alpha/2}} - 1 \right), \frac{1}{m} \left( \frac{F}{F_{\alpha/2}} - 1 \right) \right] \quad (32.1.3)$$

is an exact  $1 - \alpha$  confidence interval for  $\gamma = \sigma_a^2/\sigma^2$ , and applying the increasing map  $\gamma \mapsto \gamma/(1 + \gamma)$  to its endpoints gives an exact  $1 - \alpha$  interval for the intraclass correlation  $\rho$ . Endpoints below zero are replaced by zero.

*Proof.* By Proposition 32.1.1(d),  $F/(1 + m\gamma) \sim F(g - 1, n - g)$ , so  $\Pr\{F_{\alpha/2} \leq F/(1 + m\gamma) \leq F_{1-\alpha/2}\} = 1 - \alpha$  whatever  $\gamma$  is. Solving the two inequalities for  $\gamma$  gives Eq. 32.1.3, an event of the same probability. A strictly increasing transformation of both endpoints keeps the coverage, and  $\rho = \gamma/(1 + \gamma)$  is strictly increasing on  $[0, \infty)$ . Truncating at zero can only raise the coverage, since  $\gamma \geq 0$ . ■

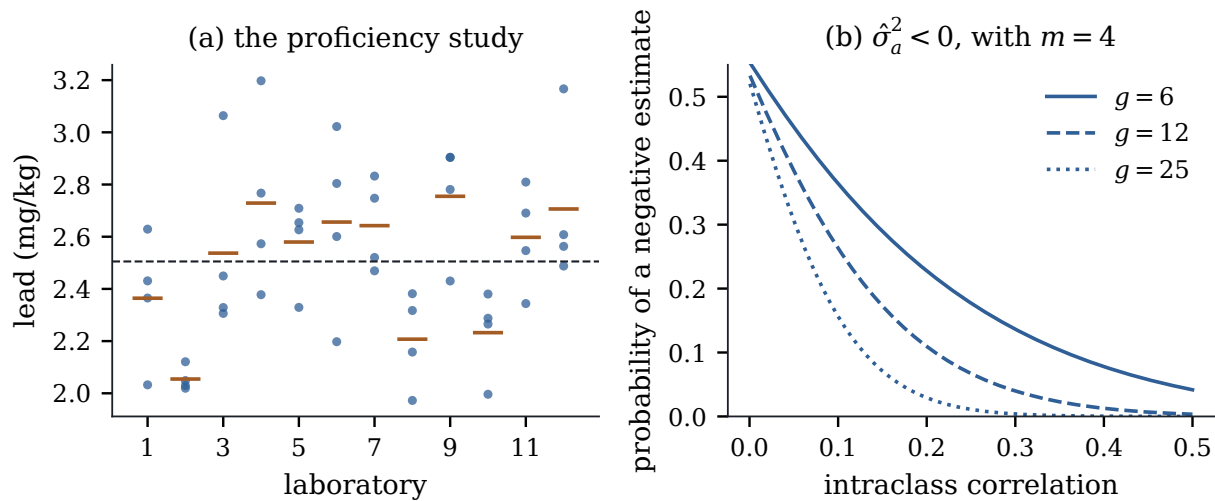


Figure 32.1.1: (a) The proficiency study: four aliquots in each of twelve laboratories, with the laboratory means (short bars) and the overall mean (dashed). (b) The probability that the moment estimate  $\hat{\sigma}_a^2$  is negative, as a function of the intraclass correlation, for  $g = 6, 12$  and  $25$  laboratories with  $m = 4$  aliquots each.

Notice what [Eq. 32.1.3](#) is not: it is an interval for the *ratio*. No exact interval of this kind exists for  $\sigma_a^2$  alone, because  $\hat{\sigma}_a^2$  is a difference of two mean squares with different scale factors and its distribution depends on both components separately. Approximate intervals for  $\sigma_a^2$  come either from Satterthwaite’s method ([Section 32.5](#)) or from the profile likelihood ([Section 32.4](#)).

#### Example. A laboratory proficiency study

Twelve laboratories each analyse four aliquots of one homogeneous batch of soil for lead, in mg/kg; the data are simulated so that the truth is known, with  $\mu = 2.50$ ,  $\sigma_a = 0.20$  and  $\sigma = 0.25$ , so that the population intraclass correlation is 0.390. The analysis of variance gives  $\text{MSB} = 0.2166$  on 11 degrees of freedom and  $\text{MSE} = 0.0620$  on 36, so

$$\hat{\sigma}^2 = 0.0620, \quad \hat{\sigma}_a^2 = \frac{0.2166 - 0.0620}{4} = 0.0386,$$

that is, a within-laboratory (repeatability) standard deviation of 0.249 and a between-laboratory standard deviation of 0.197. The estimated intraclass correlation is 0.384: about two fifths of the variation in a single measurement is attributable to which laboratory made it.

The  $F$  ratio is 3.49, and [Eq. 32.1.3](#) gives  $\gamma \in (0.118, 2.439)$ , that is  $\rho \in (0.105, 0.709)$ . Twelve laboratories and 48 measurements pin the intraclass correlation down only to a factor of seven. That is not a defect of the method; it is what 11 degrees of freedom buy.

```
import numpy as np
from scipy import stats

g, m = 12, 4                                # laboratories, aliquots each
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25       # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
```

```

y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

Y = y.reshape(g, m)                                # one row per laboratory
lab_mean, grand_mean = Y.mean(axis=1), Y.mean()

ss_between = m * np.sum((lab_mean - grand_mean) ** 2)
ss_within = np.sum((Y - lab_mean[:, None]) ** 2)
ms_between = ss_between / (g - 1)
ms_within = ss_within / (n - g)

sigma2_hat = ms_within                                # E(MS within) = sigma^2
sigma2_a_hat = (ms_between - ms_within) / m          # E(MS between) = sigma^2 + m
               sigma_a^2
icc_hat = sigma2_a_hat / (sigma2_a_hat + sigma2_hat)

print(f"MS between {ms_between:.4f}    MS within {ms_within:.4f}")
print(f"sigma^2 {sigma2_hat:.4f}    sigma_a^2 {sigma2_a_hat:.4f}    ICC {icc_hat:.3f}"
      )

```

[Run this code online](#)

```

f_obs = ms_between / ms_within                        # ~ (1 + m gamma) F(g-1, n-g)
gamma_lo = (f_obs / stats.f.ppf(0.975, g - 1, n - g) - 1) / m
gamma_hi = (f_obs / stats.f.ppf(0.025, g - 1, n - g) - 1) / m
icc_lo, icc_hi = gamma_lo / (1 + gamma_lo), gamma_hi / (1 + gamma_hi)

print(f"F = {f_obs:.2f};  gamma in ({gamma_lo:.3f}, {gamma_hi:.3f})")
print(f"intraclass correlation in ({icc_lo:.3f}, {icc_hi:.3f})")

```

[Run this code online](#)

### 32.1.3 Negative estimates

Nothing in [Proposition 32.1.1\(c\)](#) prevents MSB from falling below MSE, and then the unbiased estimate of a variance is negative. In the balanced normal model the probability is exactly

$$\Pr(\hat{\sigma}_a^2 < 0) = \Pr\left(F(g-1, n-g) < \frac{1}{1+m\gamma}\right), \quad (32.1.4)$$

by [Proposition 32.1.1\(d\)](#). Panel (b) of [Figure 32.1.1](#) plots it. With  $g = 6$  laboratories,  $m = 4$  aliquots and an intraclass correlation of 0.05 the probability is 0.453; when  $\sigma_a^2$  is exactly zero it is 0.554, a little over one half, because  $\Pr\{F(5, 18) < 1\}$  exceeds  $1/2$ . In the study of [Example \(A laboratory proficiency study\)](#), with twelve laboratories and a true correlation of 0.390, it is only 0.0143.

A negative estimate is not a computational failure but evidence, and there are three honest responses.

- (1) **Report it.** A meta-analysis that averages several studies needs the unbiased value, negative or not; truncating every study at zero and then averaging biases the average upwards.
- (2) **Truncate.** The estimate  $\max(\hat{\sigma}_a^2, 0)$  is biased upwards but has smaller mean squared error for every parameter value ([Exercise \(C1\)](#)). Maximum likelihood and REML produce it automatically, since they maximize over  $\sigma_a^2 \geq 0$ .

- (3) **Question the model.** A persistently negative estimate can mean that observations within a group are genuinely *negatively* correlated, as happens when the group is a fixed resource the observations compete for: litter-mates sharing a food supply, or plots sharing water. The matrix  $\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$  with  $\sigma_a^2 < 0$  is a perfectly good covariance as long as  $\sigma^2 + m\sigma_a^2 > 0$ , and fitting it as such is more honest than pretending a variance was estimated.

## 32.1.4 Unbalanced layouts, and beyond one way

Part (c) is the simplest instance of a general recipe: write down quadratic forms, equate them to their expectations, and solve. Henderson (1953) organized this into three methods for unbalanced layouts, of which the one used here is the first. The recipe always gives unbiased estimates and never needs normality, but it is not unique — different quadratic forms give different estimates, and in unbalanced layouts none of them is best. That is why [Section 32.4](#) turns to the likelihood.

The same calculation runs through any balanced layout. In the two-stage nested design of Exercise (C1), with  $a$  levels of  $A$ ,  $b$  of  $B$  within each and  $m$  observations per cell, that exercise gave  $E MS_{B(A)} = \sigma^2 + m\sigma_B^2$ , so the inner component is estimated by  $(MS_{B(A)} - MSE)/m$ . Taking  $A$  random too, with effects of variance  $\sigma_A^2$ ,  $MS_A$  is  $bm$  times the sample variance of  $a$  independent group means of variance  $\sigma_A^2 + \sigma_B^2/b + \sigma^2/(bm)$ , so  $E MS_A = \sigma^2 + m\sigma_B^2 + bm\sigma_A^2$  and the outer component is estimated by  $(MS_A - MS_{B(A)})/(bm)$ . Every mean square in a balanced layout turns out to estimate  $\sigma^2$  plus a positive combination of the components nested inside its factor, a fact [Chapter 33](#) establishes and uses to read a split-plot analysis off its strata.

Finally, a warning about the grand mean. In the fixed-effects model  $\text{Var}(\bar{Y}) = \sigma^2/n$ , so precision is bought with observations. Here it is different: Exercise (C2) showed that

$$\text{Var}(\bar{Y}) = \frac{\sigma^2}{n} + \sigma_a^2 \frac{\sum_k n_k^2}{n^2} \xrightarrow{\text{balanced}} \frac{\sigma^2 + m\sigma_a^2}{gm},$$

which for fixed  $g$  is bounded below by  $\sigma_a^2/g$  however large  $m$  grows. Measuring the same twelve laboratories a thousand times each does not improve the estimate of the population mean beyond a limit set by  $g = 12$ . This is the commonest way of overstating precision in clustered data, and [Chapter 33](#) quantifies it as the design effect.

## 32.1.5 Exercises

### 32.1.5.1 A. Check your understanding

#### Exercise. A1

For each study, say whether the named factor is better treated as fixed or random, and what the corresponding target of inference is. (a) Four fertilizer formulations compared on a research farm. (b) Twenty operators sampled from a factory's workforce, each measuring the same set of parts. (c) The two arms of a randomized trial. (d) Thirty schools sampled from a district, in a study of a teaching method. (e) Three furnaces, the only three the plant owns, in a study of their temperature uniformity.

### 32.1.5.2 B. Practice

#### Exercise. B1

Verify [Eq. 32.1.1](#) directly from the group means: show that  $E\{\sum_k n_k (\bar{Y}_k - \bar{Y})^2\} = \sigma^2(g -$

1) +  $\sigma_a^2(n - \sum_k n_k^2/n)$  by computing  $\text{Var}(\bar{Y}_k)$ ,  $\text{Var}(\bar{Y})$  and  $\text{Cov}(\bar{Y}_k, \bar{Y})$ . Confirm that  $m_0 = m$  when the layout is balanced, and that  $m_0 \leq (n - \min_k n_k)/(g - 1)$  in general.

*Solution.*  $\bar{Y}_k = \mu + a_k + \bar{\varepsilon}_k$  has variance  $\sigma_a^2 + \sigma^2/n_k$ , and  $\bar{Y} = \mu + \sum_k n_k a_k/n + \bar{\varepsilon}$  has variance  $\sigma_a^2 \sum_k n_k^2/n^2 + \sigma^2/n$ . Also  $\text{Cov}(\bar{Y}_k, \bar{Y}) = \sigma_a^2 n_k/n + \sigma^2/n$ . Hence  $E(\bar{Y}_k - \bar{Y})^2 = \sigma_a^2(1 - 2n_k/n + \sum_j n_j^2/n^2) + \sigma^2(1/n_k - 1/n)$ . Multiplying by  $n_k$  and summing, the  $\sigma^2$  terms give  $\sigma^2(g - 1)$  and the  $\sigma_a^2$  terms give  $\sigma_a^2(n - 2\sum_k n_k^2/n + \sum_k n_k^2/n) = \sigma_a^2(n - \sum_k n_k^2/n)$ . For the balanced case,  $n = gm$  and  $\sum_k n_k^2 = gm^2$ , so  $m_0 = (gm - m)/(g - 1) = m$ . In general  $\sum_k n_k^2 \geq n \min_k n_k$ , whence  $m_0 \leq (n - \min_k n_k)/(g - 1)$ .

### Exercise. B2

In the balanced normal model, use [Proposition 32.1.1\(d\)](#) to show that

$$\text{Var}(\hat{\sigma}_a^2) = \frac{2}{m^2} \left\{ \frac{(\sigma^2 + m\sigma_a^2)^2}{g - 1} + \frac{\sigma^4}{n - g} \right\}.$$

With  $\sigma^2$  and  $\sigma_a^2$  fixed and  $n = gm$  held constant, is the variance smaller with many small groups or few large ones?

*Solution.* MSB and MSE are independent, with  $\text{Var}(\text{MSB}) = \lambda_1^2 \cdot 2/(g - 1)$  and  $\text{Var}(\text{MSE}) = \sigma^4 \cdot 2/(n - g)$ , because a  $\chi^2(r)$  variable has variance  $2r$ . The formula follows from  $\hat{\sigma}_a^2 = (\text{MSB} - \text{MSE})/m$ . Neither extreme answers the second question. With  $g - 1 \approx n/m$  and  $n - g \approx n$ ,

$$\text{Var}(\hat{\sigma}_a^2) \approx \frac{2}{n} \left( \frac{\sigma^4}{m} + 2\sigma^2\sigma_a^2 + m\sigma_a^4 + \frac{\sigma^4}{m^2} \right),$$

a decreasing part plus an increasing one, minimized near  $m = \sigma^2/\sigma_a^2$ , that is at  $m\gamma \approx 1$ : many small groups win only once each group is large enough for MSB to stand clear of MSE. In the exact formula with  $n = 100$  and  $\sigma^2 = 1$ , at  $\sigma_a^2 = 0.01$  the variance is 0.0206, 0.0056, 0.0029 and 0.0017 for  $m = 2, 5, 10, 25$  — four groups of twenty-five beat fifty pairs by a factor of twelve — while at  $\sigma_a^2 = 1$  it is 0.1018, 0.1526 and 0.2691 for  $m = 2, 5, 10$ , and the small groups win. The extreme  $m = 1$  always fails, because then the two components are not separately identified.

### Exercise. B3

Using [Eq. 32.1.4](#), compute  $\Pr(\hat{\sigma}_a^2 < 0)$  when  $\sigma_a^2 = 0$ , for  $(g, m) = (5, 4)$  and  $(g, m) = (25, 4)$ . Explain why the probability does not tend to 0 as  $g \rightarrow \infty$  with  $\sigma_a^2 = 0$  fixed, and why it does tend to 0 for any fixed  $\sigma_a^2 > 0$ .

### 32.1.5.3 C. Going deeper

#### Exercise. C1

Let  $T$  be any estimator of  $\sigma_a^2 \geq 0$  and  $T^+ = \max(T, 0)$ . Show that  $E(T^+ - \sigma_a^2)^2 \leq E(T - \sigma_a^2)^2$ , with strict inequality whenever  $\Pr(T < 0) > 0$ . Deduce that the truncated moment estimator

dominates the unbiased one in mean squared error, and explain why this does not make the unbiased estimator useless.

*Solution.* On  $\{T \geq 0\}$  the two estimators agree. On  $\{T < 0\}$ ,  $|T^+ - \sigma_a^2| = \sigma_a^2 \leq \sigma_a^2 - T = |T - \sigma_a^2|$ , because  $T < 0$  and  $\sigma_a^2 \geq 0$ ; the inequality is strict unless  $T = 0$  almost surely on that event. Taking expectations gives the claim. The unbiased estimator remains useful because unbiasedness is preserved by averaging: combining  $K$  independent studies,  $K^{-1} \sum_k T_k$  is unbiased and consistent, whereas  $K^{-1} \sum_k T_k^+$  is biased upwards by an amount that does not vanish as  $K$  grows.

### Exercise. C2

Consider a balanced two-way layout with  $a$  levels of  $A$ ,  $b$  levels of  $B$ ,  $m$  observations per cell, and *all* effects random:  $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$  with the four sets uncorrelated and variances  $\sigma_A^2, \sigma_B^2, \sigma_{AB}^2, \sigma^2$ . Using the Kronecker projections of [Section 16.3](#) and Theorem 2.3.1, show that

$$E \text{MS}_A = \sigma^2 + m\sigma_{AB}^2 + bm\sigma_A^2, \quad E \text{MS}_{AB} = \sigma^2 + m\sigma_{AB}^2.$$

Which ratio tests  $\sigma_A^2 = 0$ ? Contrast this with the fixed-effects analysis, where  $\text{MS}_A/\text{MSE}$  is the right ratio.